Research on ADS-B Track Data Denoising Method Based on Improved Kalman Filter

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***Abstract*—To address the common noise interference problem in Automatic Dependent Surveillance-Broadcast (ADS-B) track data, this paper proposes an improved adaptive Kalman filtering algorithm. Traditional Kalman filtering is limited in performance under the conditions of ADS-B nonlinear motion models and irregular sampling. Therefore, this paper designs an optimized framework that combines feedback noise adjustment. Firstly, the geodetic latitude, longitude, and altitude of the original track data are converted into Earth Centered Fixed Coordinates (ECF) position vectors, and the ground speed, climb rate, and heading angle are converted into East North Up Coordinates (ENU) velocity vectors, which are used as equivalent measurement data. Then, spline fitting is used to remove the trend term of the equivalent measurement data, and the remaining noise errors are calibrated to obtain the statistical characteristics such as the variance of the noise. A uniform acceleration model described by Velocity Turn Climb (VTC) coordinates is constructed as the state equation, and a variable structure measurement equation matching the dimension of real-time equivalent measurement data is also established. The extended Kalman filter is introduced to smooth the ADS-B original track data. Finally, the estimated aircraft ECF position and velocity vectors are converted into standard ADS-B track data format, and the dynamic characteristics of the airborne target are calculated based on the estimated position, velocity, and acceleration data. Experimental results show that the improved algorithm has excellent noise filtering performance. This paper provides an efficient solution for improving the track data quality of the air traffic control system. Future research can further explore the integration strategy with deep learning models.**

Keywords—Improvement of Kalman filtering, Track data processing, Coordinate transformation, Noise statistics, Noise removal

I. INTRODUCTION

ADS-B, as a new type of aviation surveillance technology, has gained widespread promotion and application globally in recent years due to its excellent surveillance performance, faster data update rate, higher positioning accuracy, and lower operational costs. The International Civil Aviation Organization has identified it as one of the new technologies for future aviation surveillance development[1][2]. The trajectory information of airborne targets recorded by ADS-B is an important data source for subsequent research[3], playing a significant role in areas such as track identification, track prediction, and anomaly detection. The prerequisite for implementing these studies is high-quality ADS-B trajectory data.

However, during the actual transmission process, the ADS-B signal is subject to interference from the surrounding environment, resulting in fluctuations or even distortion of the ADS-B signal[4]. The ADS-B signal affected by noise interference has a significant impact on the decoding of the ADS-B receiver, leading to the loss and distortion of aircraft characteristic information, causing deviations between the collected ADS-B trajectory data of the airborne target and the actual position and trajectory of the airborne target[5]. These generated noise deviations have an unignorable impact on the subsequent development of airborne target recognition[6] and trajectory prediction[7]. By removing trajectory noise, it can significantly reduce trajectory deviations and improve the quality of trajectory data.

At present, many scholars have conducted in-depth research on methods for removing trajectory noise. The literature [8] focuses on the research of ship and radar track denoising and tracking in the current clutter environment, mainly covering three main lines: data association, filtering denoising, and post-processing smoothing. Regarding the problem of GPS track noise for ships[9], the main process involves using speed threshold to identify abnormal drift points + cubic spline interpolation for correction, combined with Kalman filtering for post-processing denoising of the predicted track. At the same time, a matrix neural network is used to retain temporal and spatial correlations to enhance the combined effect of denoising and prediction. The literature [10] proposes integrating deep kernel convolution into the DBSCAN-DKC method, eliminating explicit noise through kinematic rules. The literature [11] proposes an adaptive noise rate learning mechanism without prior knowledge and integrates it with JoCoR to form the A-JoCoR method. Firstly, trajectory abnormal points are eliminated through threshold and distance rules, and then the noise rate is dynamically estimated during training to adaptively screen out small loss samples. However, the current main research subjects are slow-moving targets such as ships[12], and there is relatively

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little research on noise reduction for fast-moving targets such as aircraft[13].

Track filtering can estimate the position, velocity and other state information of airborne targets[14], and update the track based on the estimated information[15], thereby eliminating noise and predicting the position information for the next moment. Track data denoising plays a crucial role in enhancing the safety and efficiency of air traffic control systems. To address the issue of noise removal in ADS-B flight track data, this paper proposes an improved Kalman filtering algorithm to optimize the noise suppression effect of ADS-B track data.

II. Mathematical Models and Problem Modeling

A. Calculate the Equivalent Measurement Data

The terrestrial longitude, terrestrial latitude and heading angle may undergo sudden changes in some scenarios, and the rate of change is usually greater than the values of the rectangular coordinate coefficients. Therefore, the conversion from geodetic latitude and longitude to the position vector in the geocentric system, and the conversion of ground speed, climb rate and heading angle to the velocity vector in the local northeast celestial system, are carried out. The geocentric position vector and the velocity vector in the northeast celestial system are defined as equivalent measurement data. The continuity of the converted data is good. The conversion method is as follows:

1) Generation of solid-state position vectors.

Let the ECF coordinates be (X, Y, Z), and in the geodetic coordinate system, the coordinates are (geodetic longitude L, geodetic latitude B, geodetic height H). The conversion from the geodetic coordinate system to the ECF system is:

$$\begin{cases} X = (N+H)\cos B\cos L \\ Y = (N+H)\cos B\sin L \\ Z = \left[N\left(1-e^2\right)+H\right]\sin B \end{cases} \tag{1}$$

In the formula, N represents the curvature radius of the ecliptic circle.

$$N = \frac{a_e}{\sqrt{1-e^2\sin^2 B}} \tag{2}$$

In the formula, a_e represents the major axis of the Earth, and e represents the eccentricity of the Earth.

2) Generation of velocity vectors for the northeastern celestial system.

Let the ground speed be v_d, the climbing rate be v_h, and the heading angle be σ. Then the velocity vector in the ENU is:

$$\boldsymbol{v}_{\text{ENU}} = \begin{bmatrix} v_d\sin\sigma \\ v_d\cos\sigma \\ v_h \end{bmatrix} \tag{3}$$

B. Measurement Data Error Calibration

The equivalent measurement data are the positions or velocities in the Cartesian coordinate system. Their continuity and smoothness are superior to the angle data. The trend term of this data time series can be obtained by using cubic spline fitting. The relationship between the measurement data sequence and the cubic spline model can be expressed in the matrix form of the measurement equation.

$$\boldsymbol{Z} = \boldsymbol{B}\boldsymbol{b} + \boldsymbol{\varepsilon} \tag{4}$$

In the formula, $\boldsymbol{Z}$ represents the measurement data sequence, $\boldsymbol{B}$ is the cubic spline model, $\boldsymbol{b}$ is the spline coefficient to be estimated, and $\boldsymbol{\varepsilon}$ is the data noise of the unknown statistical characteristic. Thus, the trajectory fitting problem is transformed into the estimation problem of the estimated parameter $\boldsymbol{b}$. The coefficient vector $\boldsymbol{b}$ is estimated using the least squares algorithm.

$$\hat{\boldsymbol{b}} = \left[\boldsymbol{B}^{\mathrm{T}}\boldsymbol{B}\right]^{-1}\boldsymbol{B}^{\mathrm{T}}\boldsymbol{Z} \tag{5}$$

The corresponding estimated covariance matrix is:

$$\boldsymbol{P}_{\hat{\boldsymbol{b}}} = \left[\boldsymbol{B}^{\mathrm{T}}\boldsymbol{B}\right]^{-1}\boldsymbol{B}^{\mathrm{T}}\boldsymbol{R}_{\varepsilon}\boldsymbol{B}\left[\boldsymbol{B}^{\mathrm{T}}\boldsymbol{B}\right]^{-1} \tag{6}$$

The estimation of equivalent measurement data is:

$$\hat{\boldsymbol{Z}} = \boldsymbol{B}\hat{\boldsymbol{b}} \tag{7}$$

Define the spline fitting error as:

$$J = \left[\boldsymbol{Z}-\hat{\boldsymbol{Z}}\right]^{\mathrm{T}}\left[\boldsymbol{Z}-\hat{\boldsymbol{Z}}\right] \tag{8}$$

For any continuous curve, the accuracy of spline fitting is closely related to the density of node placement. As the number of nodes increases from 1, the fitting error decreases rapidly with the increase in the number of nodes, and the agreement between the reconstructed curve and the measured data significantly improves; when the number of nodes continues to increase to a certain threshold, affected by the noise disturbance in the measured data, the fitting error gradually converges and approaches a constant value. Based on this, the minimum number of nodes that results in stable error can be determined as the optimal node configuration.

The specific implementation idea is: starting with the number of nodes $k=1$ as the initial condition, construct the corresponding spline fitting model and calculate the fitting error successively; increment the number of nodes one by one, compare the difference in fitting errors between nodes $k+1$ and k; when the reduction in error corresponding to adjacent node numbers is **less than the preset threshold**, this number of nodes is the optimal number of nodes that meets the accuracy requirements.

III. Adaptive Variable Structure Track Filtering Algorithm

A. State Equation

In order to facilitate the understanding of the motion characteristics of airborne targets, the state vector $\boldsymbol{X}$ is defined as consisting of the position $\boldsymbol{r}_{\text{ECF}} = \begin{bmatrix} X & Y & Z \end{bmatrix}^{\mathrm{T}}$ in the ground-fixed coordinate system, the velocity $\boldsymbol{v}_{\text{ECF}} = \begin{bmatrix} \dot{X} & \dot{Y} & \dot{Z} \end{bmatrix}^{\mathrm{T}}$, and the acceleration

$\boldsymbol{a}_{\mathrm{VTC}} = \begin{bmatrix} a_{\mathrm{V}} & a_{\mathrm{T}} & a_{\mathrm{C}} \end{bmatrix}^{\mathrm{T}}$ in the VTC coordinate system, that is:

$$\boldsymbol{X} = \begin{bmatrix} X & Y & Z & \dot{X} & \dot{Y} & \dot{Z} & a_V & a_T & a_C \end{bmatrix}^{\mathrm{T}} \quad (9)$$

Suppose that within a measurement interval, the change of the state vector is a process with a constant acceleration in the VTC system. Then, the discretized state equation is:

$$\begin{bmatrix} \boldsymbol{r}_{\mathrm{ECF}} \\ \boldsymbol{v}_{\mathrm{ECF}} \\ \boldsymbol{a}_{\mathrm{VTC}} \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{I}_{3\times3} & T\boldsymbol{I}_{3\times3} & T^2\boldsymbol{I}_{3\times3}/2 \\ \boldsymbol{O}_{3\times3} & \boldsymbol{I}_{3\times3} & T\boldsymbol{I}_{3\times3} \\ \boldsymbol{O}_{3\times3} & \boldsymbol{O}_{3\times3} & \left[\boldsymbol{T}_{\mathrm{VTC}}^{\mathrm{ECF}}\right]^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{r}_{\mathrm{ECF}} \\ \boldsymbol{v}_{\mathrm{ECF}} \\ \boldsymbol{T}_{\mathrm{VTC}}^{\mathrm{ECF}}\boldsymbol{a}_{\mathrm{VTC}} \end{bmatrix}_k + \boldsymbol{w}_k \quad (10)$$

In the formula, $\boldsymbol{T}_{\mathrm{VTC}}^{\mathrm{ECF}} = \begin{bmatrix} \dfrac{\boldsymbol{v}_{\mathrm{ECF}}}{v_{\mathrm{ECF}}} & \dfrac{\boldsymbol{r}_{\mathrm{ECF}} \times \boldsymbol{v}_{\mathrm{ECF}}}{\left\|\boldsymbol{r}_{\mathrm{ECF}} \times \boldsymbol{v}_{\mathrm{ECF}}\right\|} & \dfrac{\boldsymbol{v}_{\mathrm{ECF}}}{v_{\mathrm{ECF}}} \times \dfrac{\boldsymbol{r}_{\mathrm{ECF}} \times \boldsymbol{v}_{\mathrm{ECF}}}{\left\|\boldsymbol{r}_{\mathrm{ECF}} \times \boldsymbol{v}_{\mathrm{ECF}}\right\|} \end{bmatrix}$ represents the coordinate transformation matrix from the VTC system to the ground fixed system, T is the measurement interval, $\boldsymbol{I}_{n\times n}$ is an n-order identity matrix, $\boldsymbol{O}_{m\times n}$ is an m-row n-column zero matrix, $\boldsymbol{w}_k$ represents the process noise, and is an unknown zero-mean Gaussian white noise.

B. Variable Structure Measurement Equation

The measurement equation needs to provide the functional relationship between the state vector and the measurement vector. For ADS-B data, the original measurement data includes geodetic longitude, geodetic latitude, geodetic height, ground speed, climb rate, and heading angle. Given these six characteristic components, the equivalent measurement data are the x, y, and z components of the position vector in the earth-fixed system and the velocity vector in the northeast celestial system. At this point, the measurement vector and the measurement equation are

$$\boldsymbol{y}_6 = \begin{bmatrix} X \\ Y \\ Z \\ v_x \\ v_y \\ v_z \end{bmatrix}, \quad \begin{aligned} &\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \boldsymbol{H}_1\boldsymbol{X} \\ &\boldsymbol{v}_{\mathrm{ENU}} = \boldsymbol{T}_{\mathrm{ECF}}^{\mathrm{ENU}}\boldsymbol{v}_{\mathrm{ECF}} \\ &v_x = \boldsymbol{v}_{\mathrm{ENU}}(1) \\ &v_y = \boldsymbol{v}_{\mathrm{ENU}}(2) \\ &v_z = v_h = \boldsymbol{v}_{\mathrm{ENU}}(3) \end{aligned} \quad (11)$$

Based on the actual received ADS-B flight track data, there are missing values for ground speed, climb rate and heading angle. Therefore, the dimensions and specific expressions of the measurement equation need to change according to the content of the measurement data. When only the longitude, latitude and altitude are known, the equivalent measurement data is the ground-fixed position vector. At this time, the measurement vector and the measurement equation are as follows:

$$\boldsymbol{y}_1 = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \boldsymbol{H}_1\boldsymbol{X} \quad (12)$$

In the equation $\boldsymbol{H}_1 = \begin{bmatrix} \boldsymbol{I}_{3\times3} & \boldsymbol{O}_{3\times3} & \boldsymbol{O}_{3\times3} \end{bmatrix}$.

C. Extended Kalman Filter

After obtaining the state estimate $\hat{\boldsymbol{X}}_{k/k}$ and covariance estimate $\hat{\boldsymbol{P}}_{k/k}$ at time k, the algorithm flow of the extended Kalman filter is as follows:

1) State prediction.

$$\hat{\boldsymbol{X}}_{k+1/k} = \boldsymbol{f}_k\left(\hat{\boldsymbol{X}}_{k/k}\right) \quad (13)$$

In the equation, the function $\boldsymbol{f}_k(\cdot)$ is the state prediction formula (10).

2) Covariance prediction.

$$\hat{\boldsymbol{P}}_{k+1/k} = \boldsymbol{\Phi}\hat{\boldsymbol{P}}_{k/k}\boldsymbol{\Phi}^T + \boldsymbol{Q}_{k+1} \quad (14)$$

In the formula, $\boldsymbol{\Phi} = \left.\dfrac{\partial \boldsymbol{f}_k}{\partial \boldsymbol{X}}\right|_{\boldsymbol{X}=\hat{\boldsymbol{X}}_{k+1/k}}$ represents the state transition matrix, and B is the covariance matrix of process noise, which is set to

$$\boldsymbol{Q}_{k+1} = \frac{Ta_{\max}^2}{6\tau_m}\begin{bmatrix} \boldsymbol{O}_{6\times6} & \boldsymbol{O}_{3\times6} \\ \boldsymbol{O}_{6\times3} & \boldsymbol{I}_{3\times3} \end{bmatrix} \quad (15)$$

In the formula, τ_m represents the maneuvering time constant, and $a_{\max}$ represents the maximum acceleration of the target.

3) Gain matrix calculation.

$$\boldsymbol{K} = \hat{\boldsymbol{P}}_{k+1|k}\boldsymbol{H}_i^{\mathrm{T}}\left[\boldsymbol{H}_i\hat{\boldsymbol{P}}_{k+1|k}\boldsymbol{H}_i^{\mathrm{T}} + \boldsymbol{R}_{k+1}\right]^{-1} \quad (16)$$

4) State estimation update.

$$\hat{\boldsymbol{X}}_{k+1/k+1} = \hat{\boldsymbol{X}}_{k+1/k} + \boldsymbol{K}\left[\boldsymbol{y}_{k+1} - \boldsymbol{h}_i\left(\hat{\boldsymbol{X}}_{k+1/k}\right)\right] \quad (17)$$

5) Covariance update.

$$\hat{\boldsymbol{P}}_{k+1/k+1} = \left[\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H}_i\right]\hat{\boldsymbol{P}}_{k+1/k} \quad (18)$$

For the extended Kalman filter, since both the state equation and the measurement are nonlinear, the state transition matrix and the observation matrix can be solved using numerical methods.

IV. Experimental Design and Result Analysis

A. Experimental Design

Simulate the complete flight process of a certain aircraft, and convert the standard flight path data into geodetic coordinates for position and velocity. The standard flight path is shown in the figure.

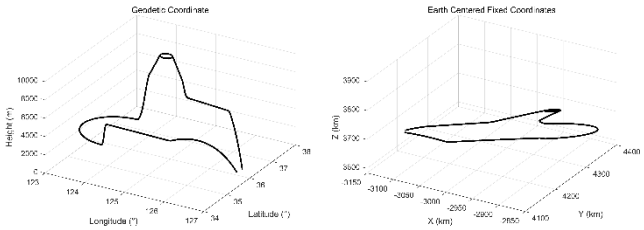

Fig. 1. Standard three-dimensional trajectory.

B. Result Analysis

Assuming the ground solid position is the equivalent measured data, and by superimposing white noise with a mean of 0 and a standard deviation of 100m to simulate the actual noisy measured data, using the filtering algorithm proposed in this paper, the measured data is smoothed and filtered. Monte Carlo simulation is conducted 500 times, and the root mean square error of the position estimation is statistically calculated to evaluate the estimation accuracy and effectiveness of the filtering algorithm. The position estimation error is shown in the figure:

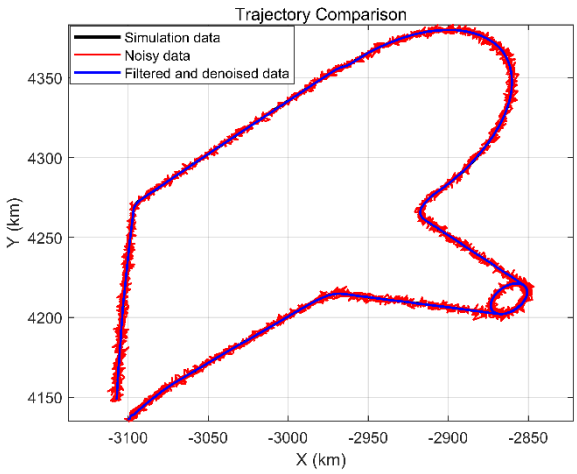

Fig. 2. Trajectory comparison.

The position and speed estimation results are shown in the table.

TABLE I. FILTERING ERROR ASSESSMENT

Variable	RMSE
X-axis	64.37 m
Y-axis	50.06 m
Z-axis	63.02 m
Ground speed	0.93 m/s

From the results, it can be seen that when the filtering reaches a steady state, the root mean square error of the position estimation is approximately 59.15 meters, which represents a certain improvement in accuracy compared to the 100-meter standard deviation of the original measurement data.

V. CLOSING REMARKS

For the problem of noise removal in ADS-B trajectory data of airborne targets, an improved adaptive Kalman filtering algorithm is proposed. Using noisy ADS-B trajectory data, the validity of the model and algorithm is verified. The algorithm has low time complexity and fast computing speed, and can meet the requirements of different data processing scenarios. In the future, this method will be used to conduct research on the actual application scenarios of online noise removal.

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